

COURSE: CHE 312: Transport Phenomena II
Spring Semester 2020
Department of Chemical Engineering
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Quiz #4

April 17, 2020

Online

Duration: 25 minutes

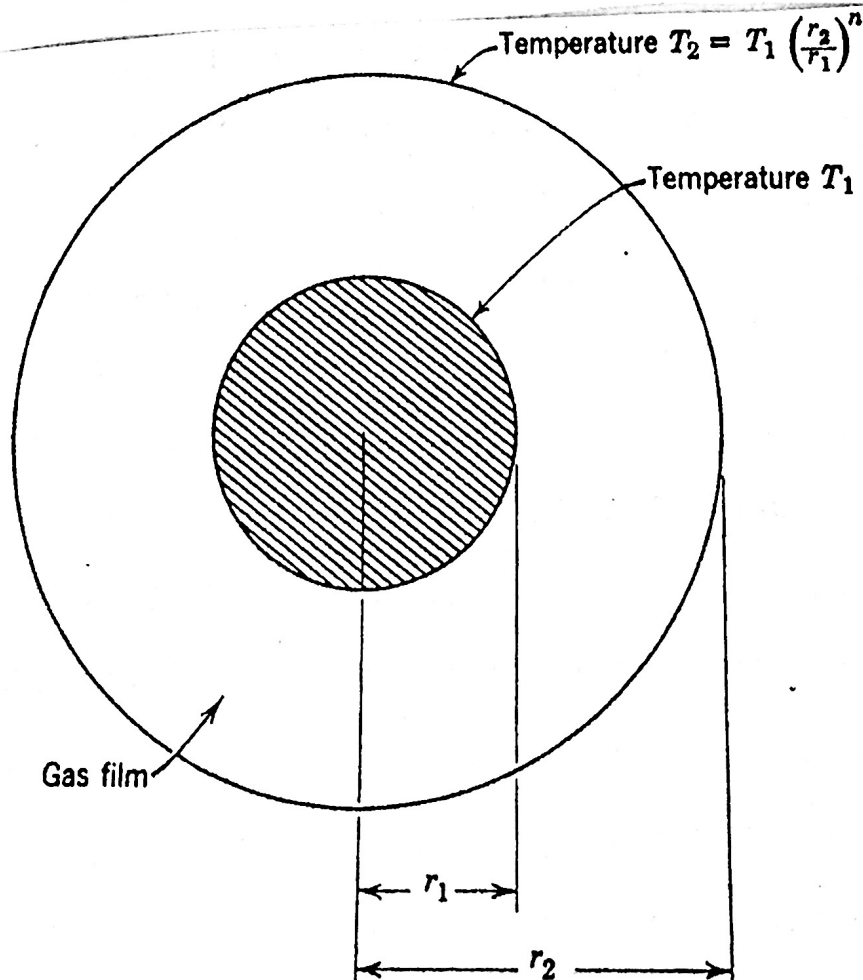
Describe the diffusion in a non-isothermal film in which the temperature changes with the distance according to:

$$\left(\frac{T}{T_1}\right) = \left(\frac{r}{r_1}\right)^n$$

in which T_1 is the temperature at $r = r_1$. Assume as a rough approximation that D_{AB} varies as the $3/2$ - power of the temperature:

$$\left(\frac{D_{AB}}{D_{AB,1}}\right) = \left(\frac{T}{T_1}\right)^{3/2}$$

In which $D_{AB,1}$ is the diffusivity at $T = T_1$. Problems like this arise in connection with drying droplets and diffusion through gas films near spherical catalysts pellets.



Diffusion through a spherical film